Lesson End Project

Designing a Network Solution

Project agenda: To design a network solution

Description: You are working as a principal engineer in an organization, and you have been asked to design a solution for an on-premises network to deploy a virtual appliance. The plan is to deploy multiple Azure VMs and connect the on-premises network to Azure using a site-to-site connection. You need to also ensure that all the network traffic that will be directed from the Azure VMs to a specific subnet must flow through the virtual appliance. As part of compliance, the VMs should not access the internet as they store sensitive data. Recommend a design considering the above requirements.

Tools required: Azure account with administrator credentials

Prerequisites: None

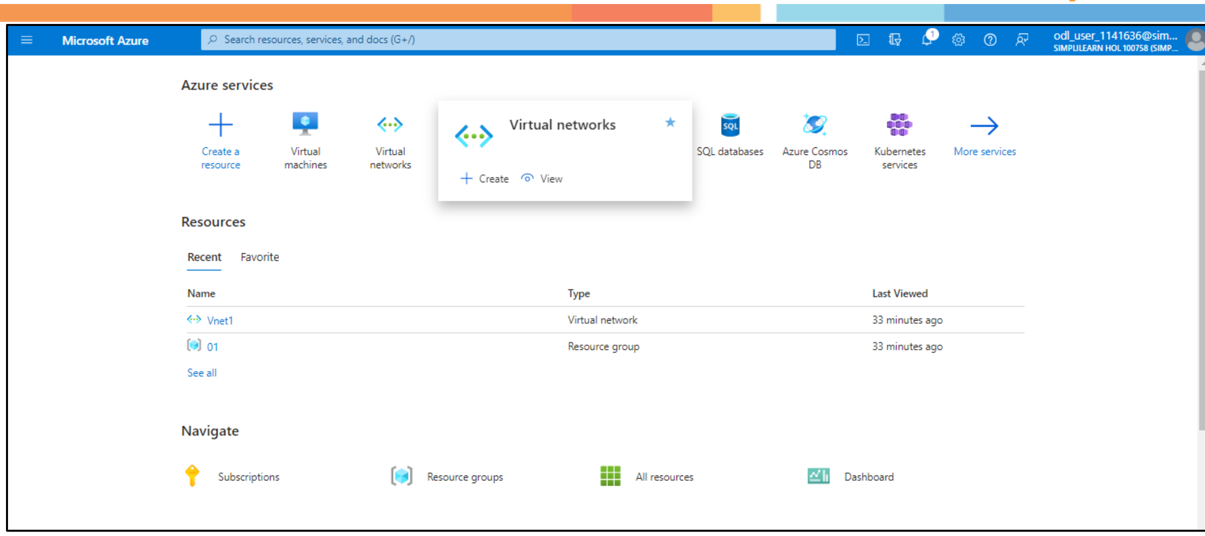
Expected deliveries: Added the screenshots for every step to design a network solution.

Steps to be followed:

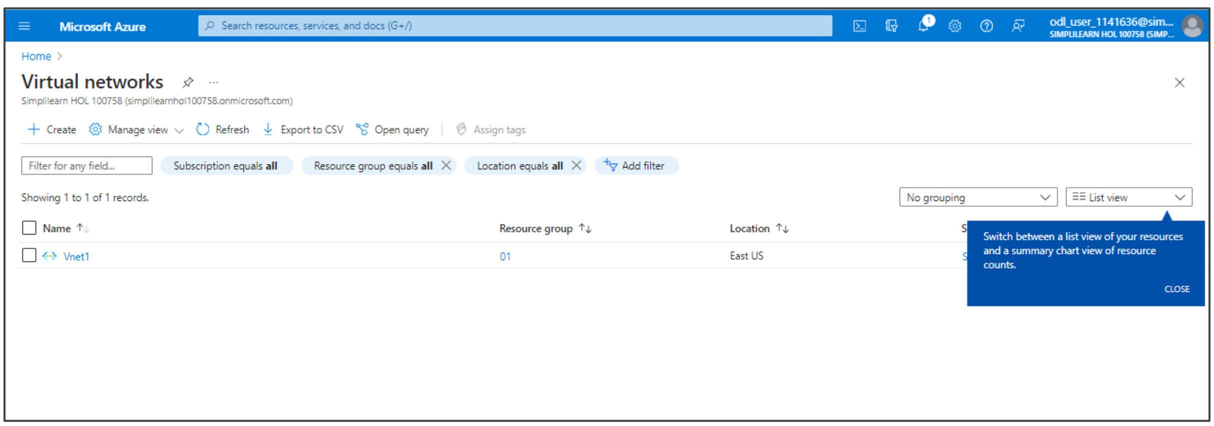
1. Create a virtual network
2. Deploy the virtual machine in the respective virtual network
3. Verify the network security group of the virtual machine

Step 1: Create a virtual network

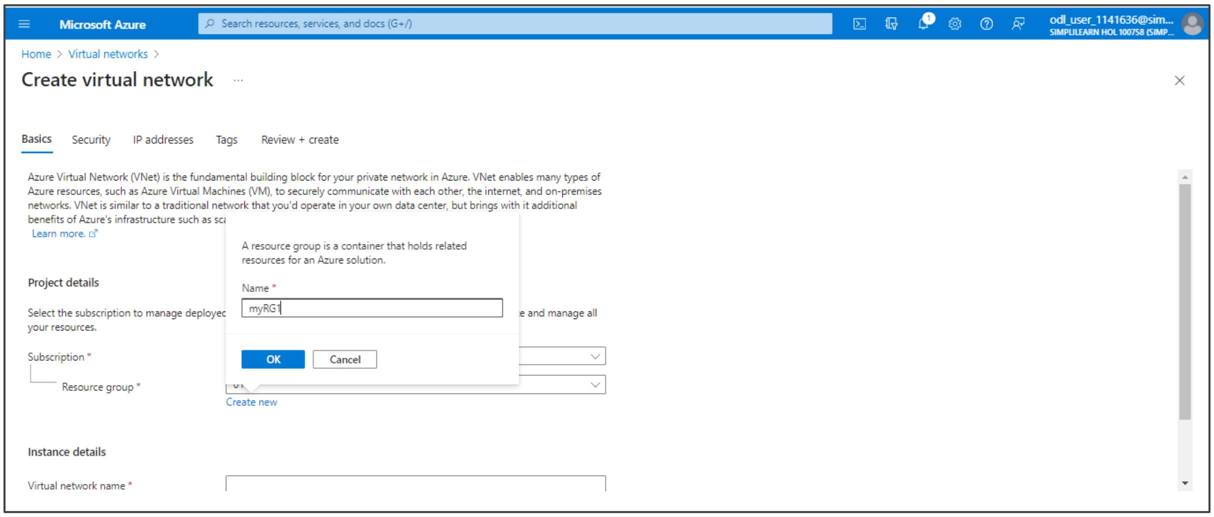
1.1 Log in to the Azure portal, search and select **Virtual networks**



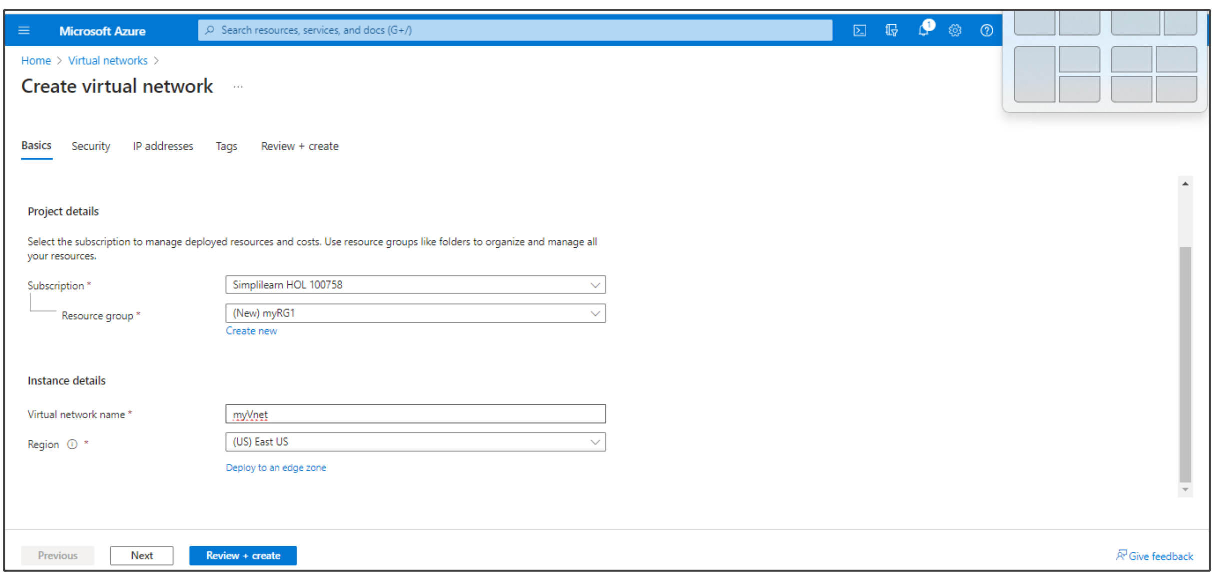
2.1 Click on the **+ Create**



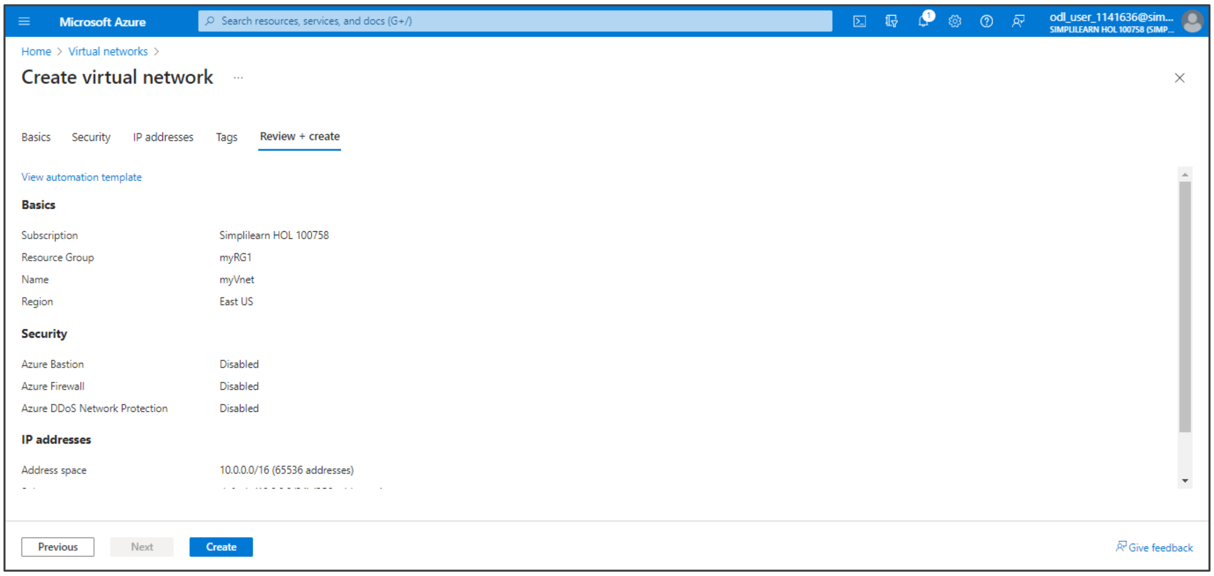
2.2 On the **Basics** page, under **Resource group** create a new resource group **(myRG1)**



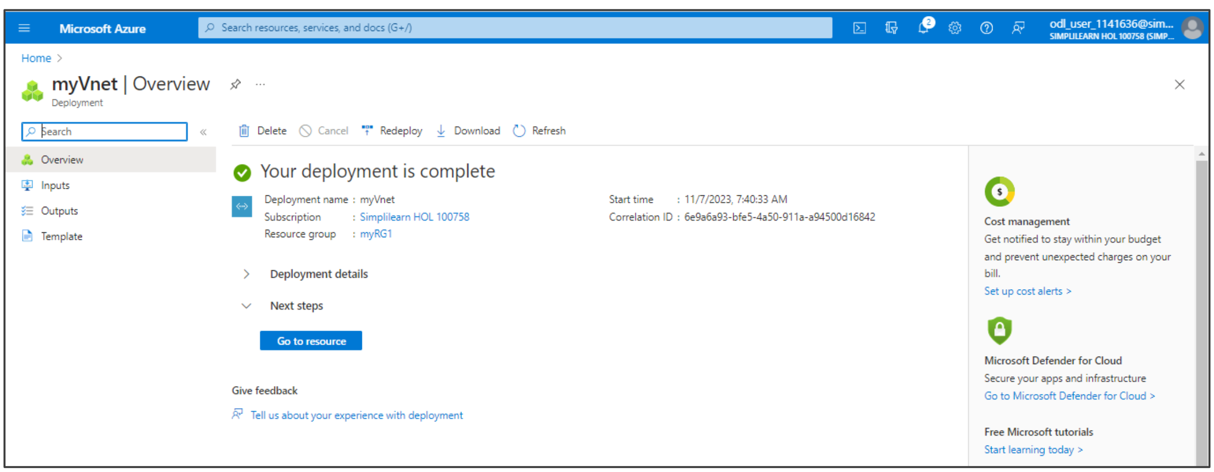
2.3 Under **Instance details**, enter **Virtual network name** as **myVnet** and click on **Review + create**



2.4 Click on the **Create**

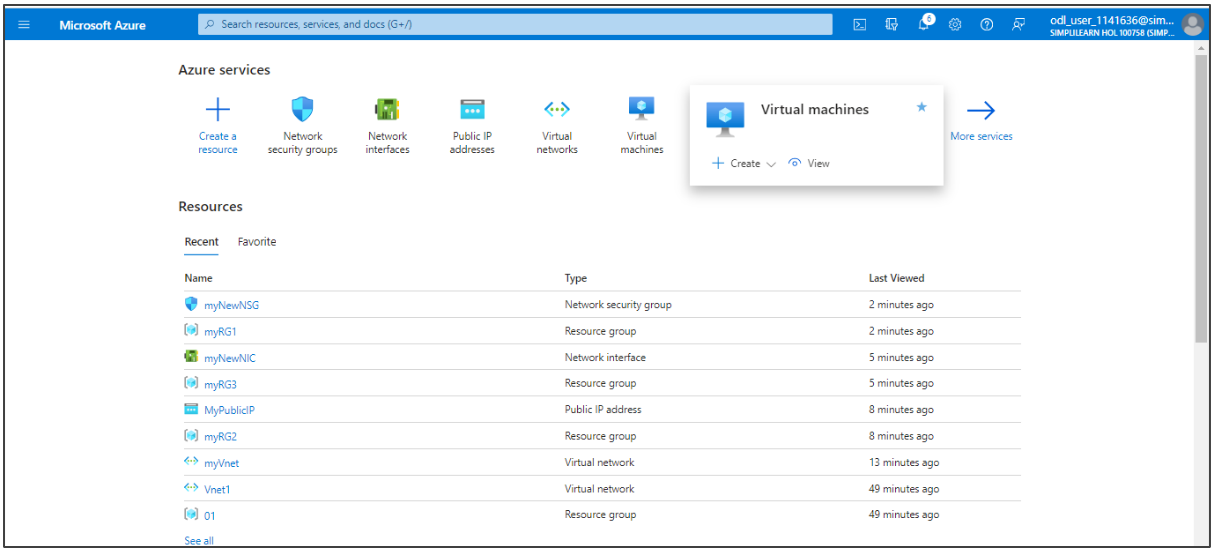


The output appears as shown below:

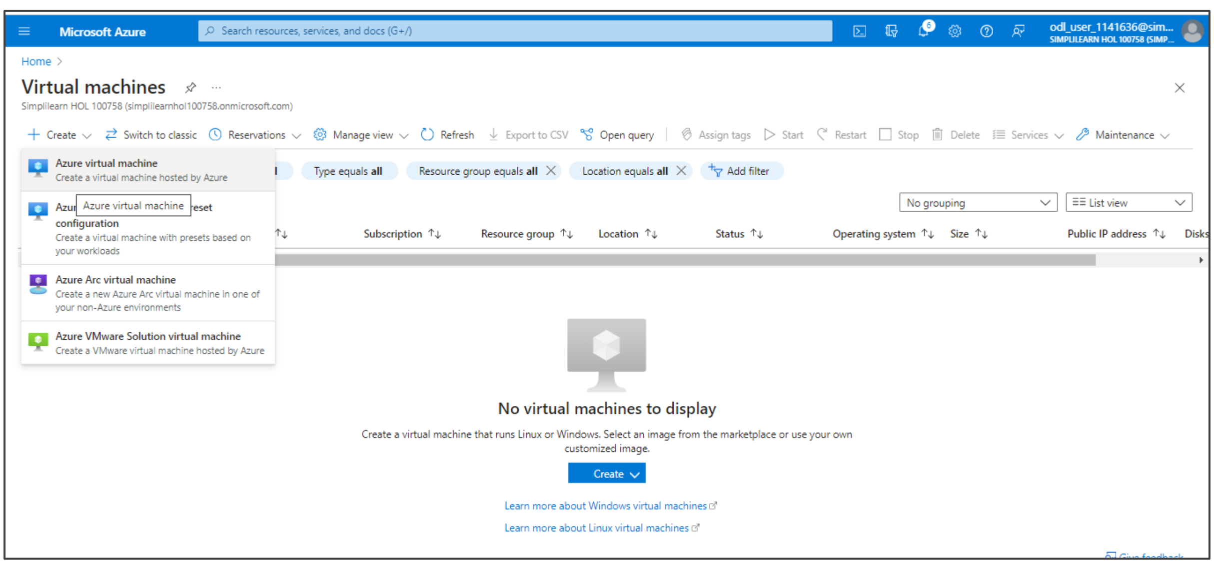


Step 2: Deploy the virtual machine in the respective virtual network

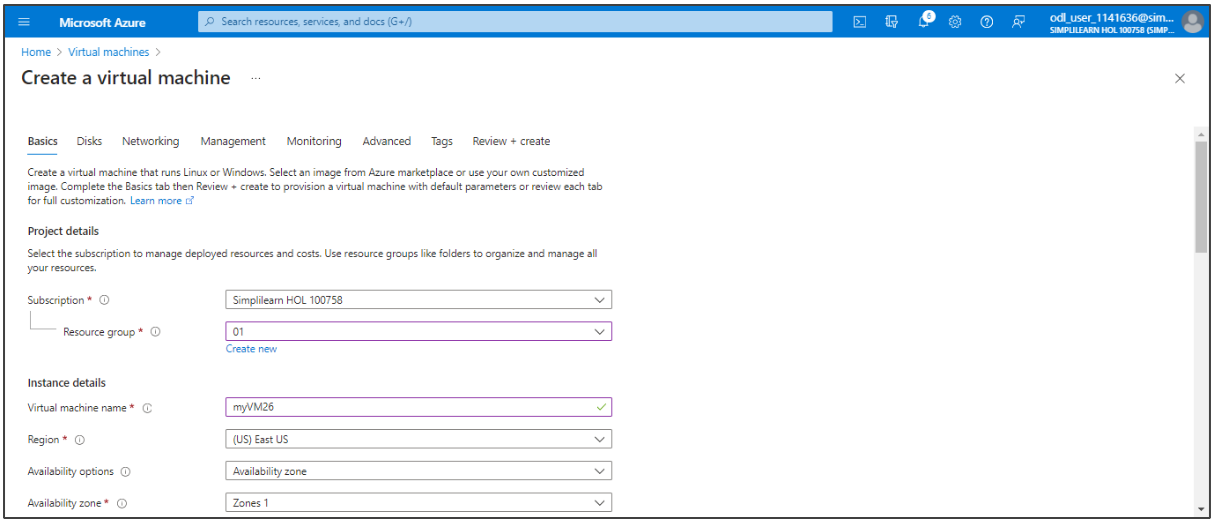
3.1 In the Azure portal, search and select **Virtual machines**

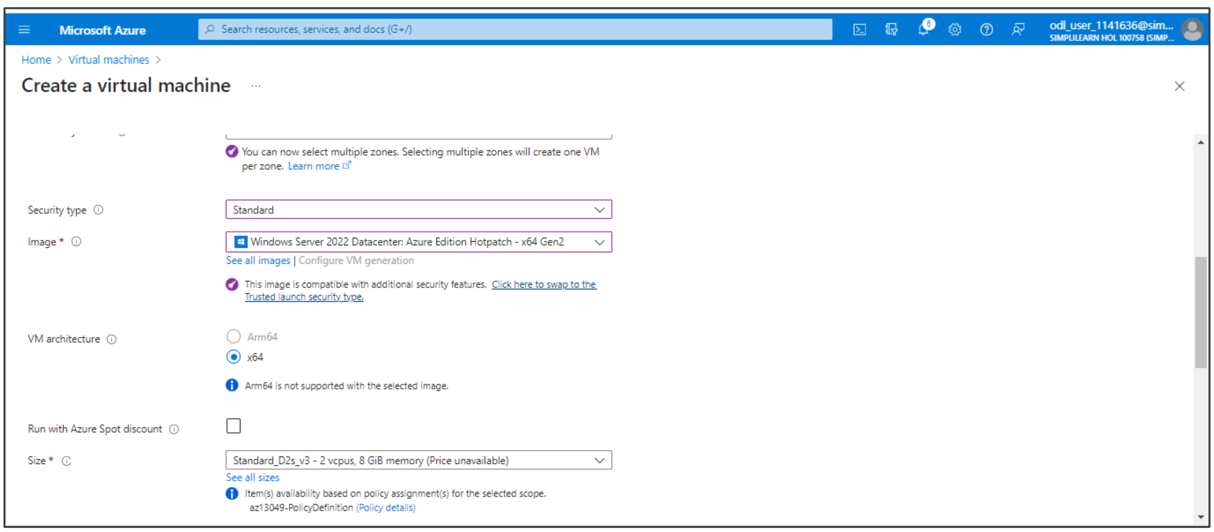


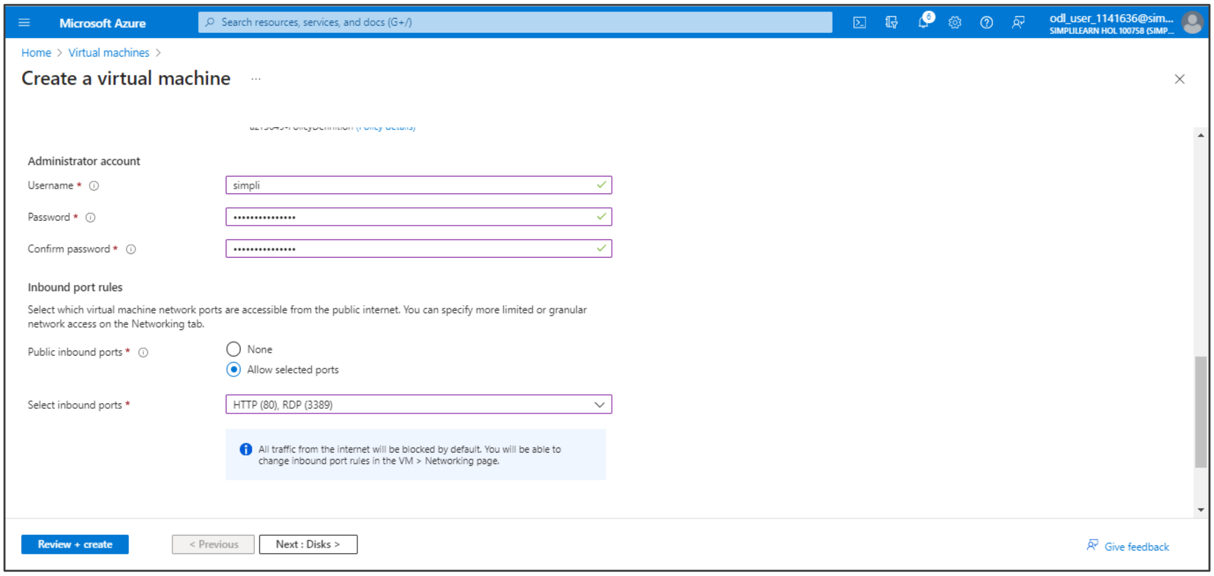
3.2 Click on **+ Create** and select **Azure virtual machine**



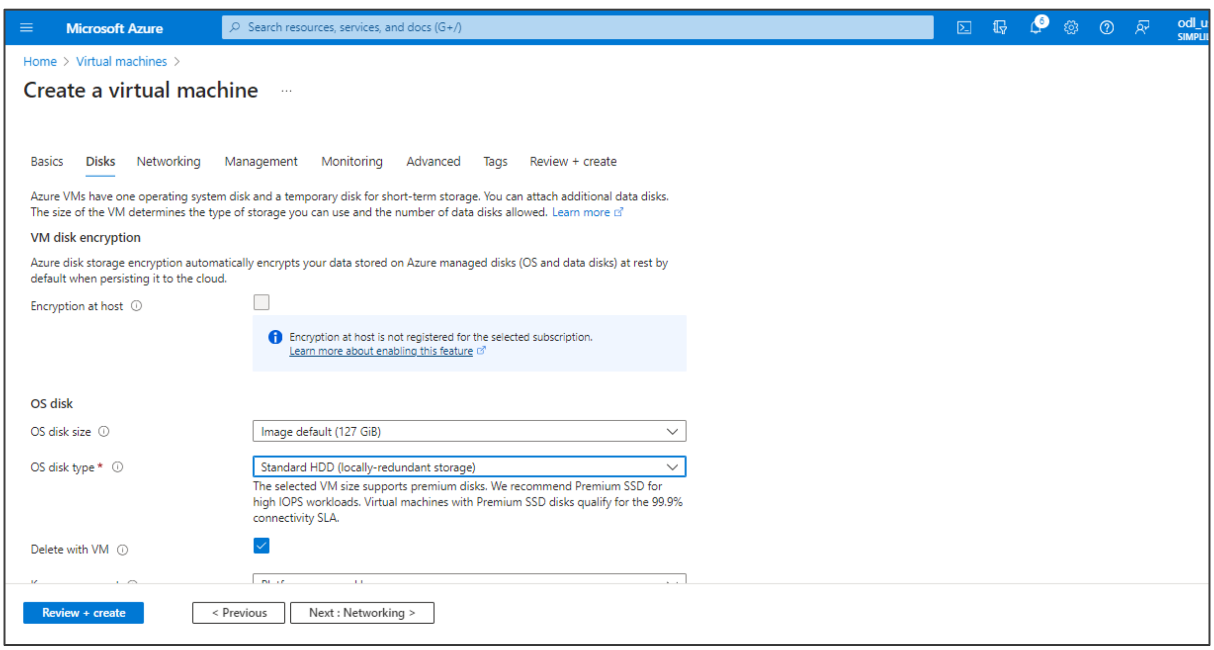
3.3 On the **Basics** page, enter the following details as shown below and click on **Next : Disks >**



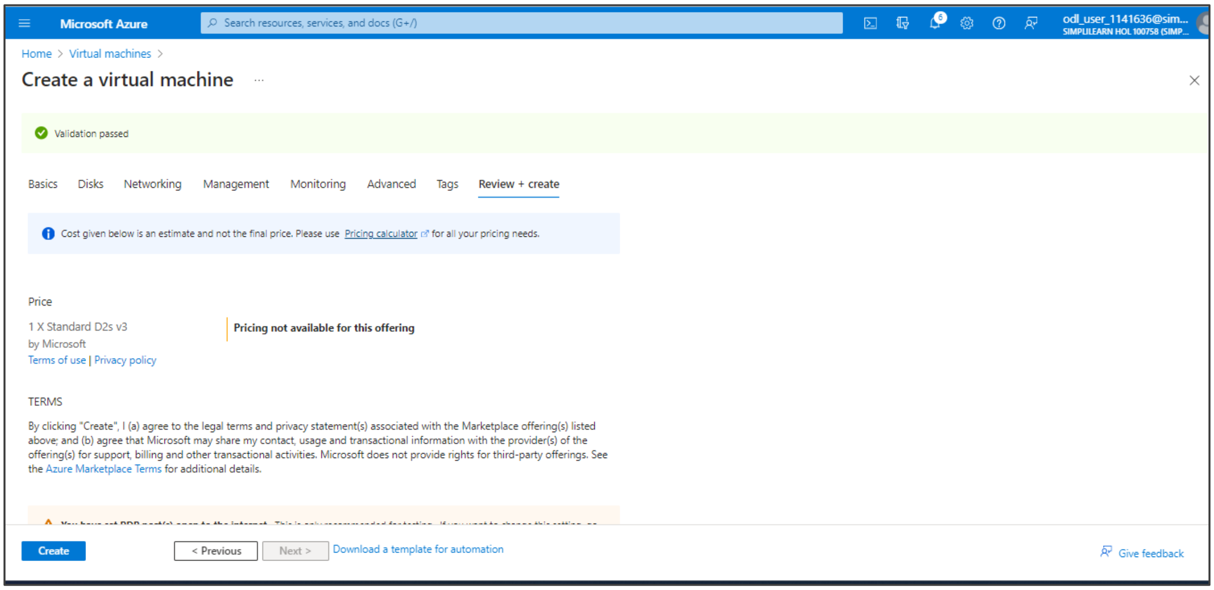




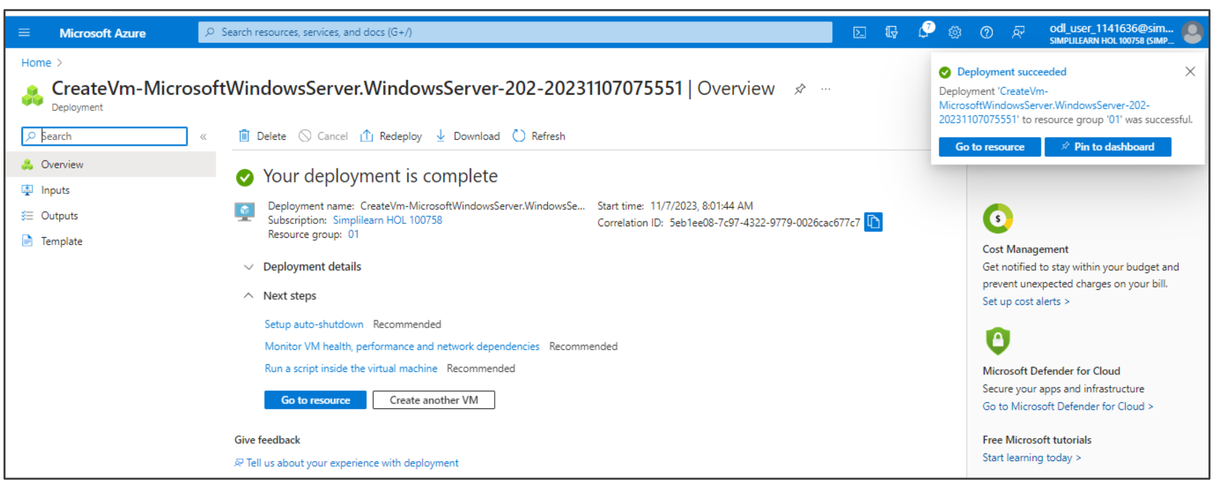
3.4 On the **OS disk** page, under **OS disk type** select **Standard HDD (locally-redundant storage)** and click on **Review + create**



3.5 Click on the **Create**

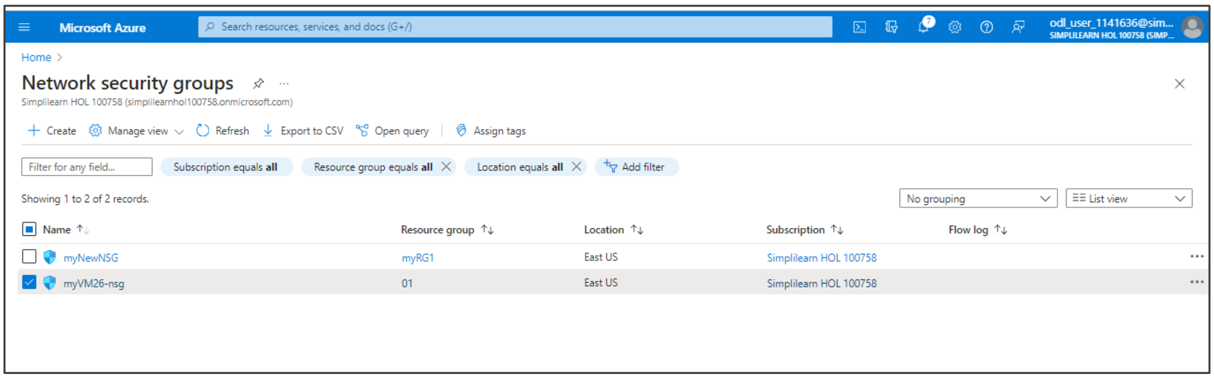


The output appears as shown below:

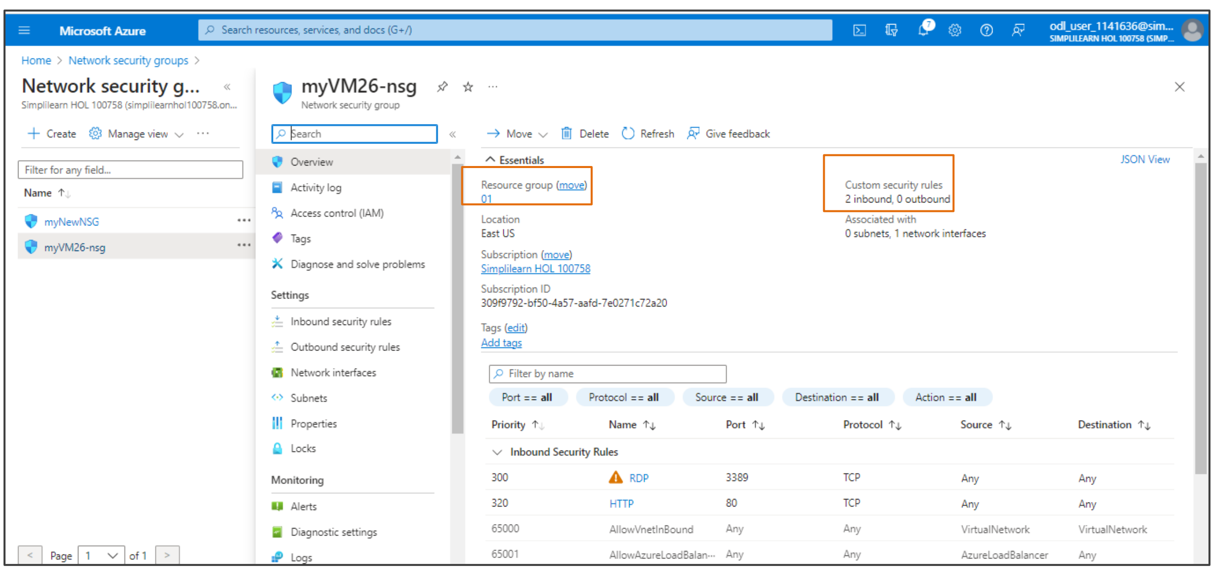


Step 4: Verify the network security group of the virtual machine

4.1 After the deployment of the virtual machine using the given virtual network under the respective resource group. The user can verify the NSG (network security group) of the virtual machine



4.2 The user can verify the network security group (NSG) associated with the VM. The NSG is crucial for controlling inbound and outbound traffic to the VM.



By following these steps, you have successfully created a network design that integrates on-premise infrastructure with Azure through a site-to-site connection, deploying multiple VMs.